

Cell-surface antigen discovery in lung adenocarcinoma

Antibody-accessible ADC / CAR-T target nomination from tumour single-cell expression, extracellular topology and normal-tissue therapeutic index.

Generated 2026-09-16 from CZ CELLxGENE Census 2025-11-08. All figures in this report are read back from the result files on disk.

Design principle

Targets are ranked by **tumour surface specificity x normal-tissue therapeutic index**, not by tumour essentiality. An antibody therapeutic kills cells that display the antigen; whether the tumour needs the gene to survive is irrelevant to efficacy but directly relevant to toxicity, because an essential gene is essential in normal tissue too. DepMap gene effect is therefore carried as annotation and never as a filter.

1. Search space

The search space is the SURFY in-silico surfaceome. Of 20,193 proteins in the master table, 2,886 carry a surface label, collapsing to **2,799 unique genes**: 1,254 with experimental evidence (CSPA peptide capture or a GPI anchor) and 1,545 from the machine-learning prediction. Each gene keeps its own topology string, so extracellular domain length and transmembrane pass count are parsed per gene rather than assumed. The pre-registered validation set was locked at this point: 9/9 clinically validated LUAD antigens and 3/4 negative controls fall inside the search space.

2. Tumour single-cell expression

CZ CELLxGENE Census 2025-11-08 was queried for primary 'lung adenocarcinoma' cells. Single-nucleus data systematically under-detects membrane-protein transcripts and was excluded, leaving **831,387 whole cells across 5 atlases**. Each atlas was sampled to at most 20,000 cells with seed 0 (82,558 cells), and 75,095 of those carry a cell-type label that maps to one of the four compartments: caf 2,270, endothelial 2,544, epithelial 12,853, immune 57,428. Per-gene mean expression and detection rate were computed per compartment per atlas, then integrated into a cross-atlas consensus.

3. Extracellular topology gate

Proteins that cannot present an epitope on an intact cell were removed: **174 genes excluded**, 104 on subcellular localisation (ER, Golgi, endosome, nucleus, cytoplasm or purely secreted / matrix) and 70 for having no extracellular segment of at least 25 residues. **2,625 antibody-accessible candidates** remain, each labelled with its surface-confirmation tier.

4. Druggability and essentiality annotation

Open Targets supplied the antibody tractability bucket, an independent subcellular localisation call and the drug record for 2,361 candidates. Surface confirmation splits as: Open Targets confirmed 868; experimental (CSPA/GPI) 1,217; unconfirmed (ML prediction only) 540. DepMap mean gene effect was attached for 18,443 genes as annotation only. The validated antigens confirm why: CD276 -0.26, CEACAM5 -0.03, EGFR -0.35, ERBB2 -0.34, FOLR1 +0.02, MET -0.12 -- all far from essential. Gating on essentiality would have removed the targets this pipeline is supposed to rediscover and enriched for housekeeping genes instead.

5. Normal-tissue safety baseline

Two independent Human Protein Atlas releases set the safety baseline: consensus RNA (20,151 genes x 51 tissues) and normal-tissue immunohistochemistry (11,152 genes x 59 tissues). Each arm gives an organ-weighted risk, and the safety coefficient is the **conservative minimum of the two**, so a target that looks clean in RNA but stains in heart muscle is treated as dangerous. 139 candidates had no HPA record and received the flagged neutral default 0.7.

Median safety coefficient across the cohort is 0.373.

6. Composite score and tiers

The composite is the **geometric mean of tumour quality, safety coefficient and consensus multiplier** -- the cube root of their product, a monotone transform that leaves every rank exactly where the raw product puts it while keeping the score on the same 0-1 scale as its three factors. Tumour quality weights relative TME specificity 0.30, expression intensity 0.20, expression uniformity 0.20, ectodomain accessibility 0.15 and antibody druggability 0.15. Safety enters multiplicatively and the consensus multiplier rewards reproducibility across atlases. One eligibility rule is applied before ranking: a candidate must be detected in at least 10% of epithelial/malignant cells, because an ADC or a CAR cannot engage an antigen that most tumour cells never display. That sets aside 2,127 genes which would otherwise win on a safety coefficient earned purely by being silent everywhere, and leaves 379 scored candidates (119 accessible genes are absent from the Census feature set altogether). Absolute thresholds give **3 Tier 1** (≥ 0.55), 76 Tier 2 (0.35-0.55) and 300 Tier 3 targets.

Cohort

atlas	cells available	sampled	analysed	donors	assay
Lung cancer 4 patients archival FFPE samples profiling with Chromium FLEX	2,558	2,558	2,454	2	10x gene expression flex
The single-cell lung cancer atlas (LuCA) -- extended atlas	647,147	20,000	19,910	154	10x 3' v1, 10x 3' v2, 10x 3' v3, 10x 5' v1, BD Rhapsody Whole Transcriptome Analysis, GEXSCOPE technology, Smart-seq2, inDrop
Combined samples	57,679	20,000	20,000	21	10x 3' transcription profiling, 10x 3' v2, 10x 3' v3
An integrated cell atlas of the human lung in health and disease (full)	41,012	20,000	12,731	3	10x 3' v2
Transcriptional connectivity of regulatory T cells in the tumor microenvironment informs novel combination cancer therapy strategies	82,991	20,000	20,000	23	10x 3' v2, 10x 3' v3

7. Validation and quality control

recall@10 = 2/9, recall@20 = 3/9 for the pre-registered clinically validated antigens among 379 scored candidates. The best-placed validated antigen is CEACAM5 at rank 7.

The validated antigens are not buried, and reading the recall number alone would be misleading. 8 of 9 were ranked at all, between rank 7 and 126, with a median position in the top 13.7% of scored candidates. On the tumour side they are clearly above the field: median tumour quality 0.630 against a cohort median of 0.306. Their safety coefficients are also above the cohort median (0.131 against 0.050).

What keeps them out of the top ten is that the leaders are cleaner in normal tissue, not that the validated antigens look bad: the top ten carry a median safety coefficient of 0.337 at a median tumour quality of 0.509. EGFR, HER2, TROP2 and folate receptor alpha are genuinely expressed in skin, gastrointestinal tract, kidney and heart, and their clinical use is bought with managed toxicity, dose reduction and patient selection. A purely expression-driven safety coefficient prices that cost in; a clinical programme can decide to pay it. So the honest reading of this ranking is not that it failed to rediscover the standard of care, but that it is answering a different question: which antigens have a *better* normal-tissue profile than the targets already in the clinic. A ranking retuned until the known answers came back on top would be a ranking tuned to its own validation set.

Not ranked at all: ROR1 (displayed on 4.6% of epithelial cells, below the 10% antigen display threshold).

Pre-registered validated antigens

gene	rank	percentile	score	tumour quality	safety	worst normal tissue	note
CEACAM5	7	top 1.6%	0.528	0.701	0.300	Colon	
MSLN	9	top 2.1%	0.515	0.777	0.175	Colon	
EGFR	15	top 3.7%	0.500	0.498	0.279	Kidney	
CD276	45	top 11.6%	0.389	0.315	0.373	Liver	
ERBB2	59	top 15.3%	0.369	0.585	0.086	Heart muscle	
FOLR1	98	top 25.6%	0.329	0.710	0.050	no HPA record	
TACSTD2	107	top 28.0%	0.323	0.676	0.050	Kidney	
MET	126	top 33.0%	0.307	0.577	0.050	Cerebral cortex	
ROR1	-	-	n/a	n/a	n/a	no HPA record	displayed on 4.6% of epithelial cells, below the 10% antigen display threshold

Negative controls: EPCAM PASS (ranked 83 of 379); CDH1 PASS (ranked 99 of 379); ATP1A1 PASS (ranked 238 of 379); CGN PASS (never entered the search space (SURFY calls it non-surface)). Overall gate: **PASS**.

Rank stability across 6 alternative safety-aggregation and weighting schemes: median Spearman correlation with the baseline ranking 0.870, worst top-20 overlap 8/20. The head of the ranking is therefore not an artefact of one particular parameterisation.

alternative scheme	Spearman vs baseline	top-20 overlap	Tier 1 count
safety = mean(protein, RNA) instead of the conservative min	0.801	15/20	9
safety = IHC protein arm only	0.658	8/20	33
safety = consensus RNA arm only	0.687	13/20	6
safety enters as sqrt(safety): a softer toxicity penalty	0.940	17/20	31
all five tumour-quality sub-scores weighted 0.20	0.996	18/20	4
specificity 0.50, the other four 0.125 each	0.992	20/20	7

Top nominations

Each of the top 12 candidates was queried against Europe PMC with fixed query shapes for indication biology, surface-targeting modality and clinical stage. Evidence classes: preclinical surface-targeting evidence (6); clinically validated target (4); novel -- no surface-targeting literature (1); tumour-biology evidence only (1).

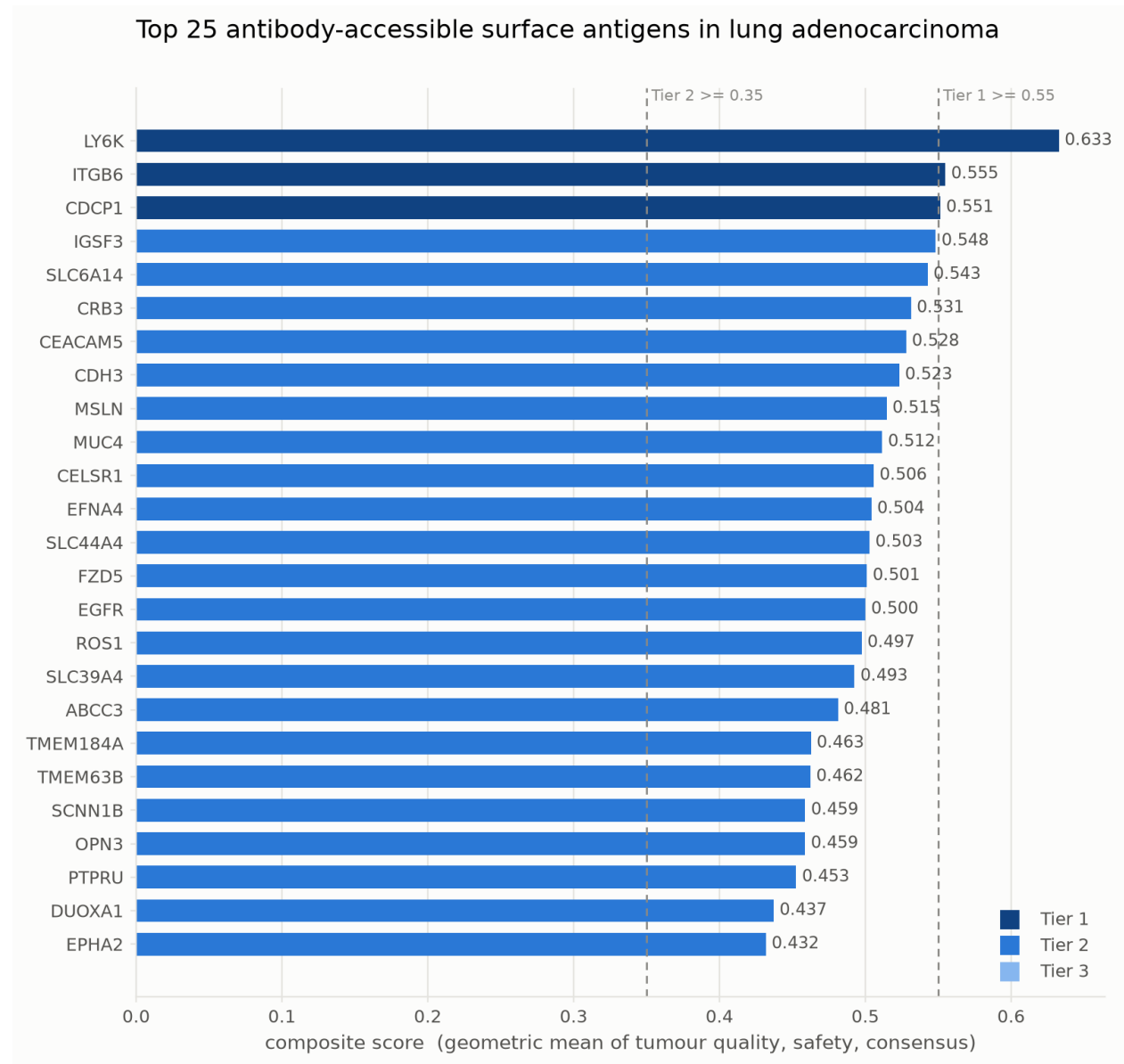
#	gene	score	tier	specificity	safety	topology	surface confirmation	evidence class
1	LY6K	0.633	Tier 1	0.77	0.60	GPI / no TM	experimental	preclinical surface-targeting evidence
2	ITGB6	0.555	Tier 1	1.00	0.24	type I	Open Targets	clinically validated target
3	CDCP1	0.551	Tier 1	0.45	0.37	type I	experimental	preclinical surface-targeting evidence
4	IGSF3	0.548	Tier 2	0.55	0.49	type I	ML prediction only	novel -- no surface-targeting literature
5	SLC6A14	0.543	Tier 2	0.78	0.46	multi-pass (12 TM)	Open Targets	tumour-biology evidence only
6	CRB3	0.531	Tier 2	0.97	0.28	type I	Open Targets	preclinical surface-targeting evidence
7	CEACAM5	0.528	Tier 2	1.00	0.30	GPI / no TM	experimental	clinically validated target
8	CDH3	0.523	Tier 2	0.52	0.37	type I	experimental	clinically validated target
9	MSLN	0.515	Tier 2	1.00	0.18	GPI / no TM	experimental	clinically validated target
10	MUC4	0.512	Tier 2	0.82	0.30	type I	experimental	preclinical surface-targeting evidence
11	CELSR1	0.506	Tier 2	0.41	0.36	multi-pass (7 TM)	experimental	preclinical surface-targeting evidence
12	EFNA4	0.504	Tier 2	0.24	0.55	GPI / no TM	experimental	preclinical surface-targeting evidence

Literature triage

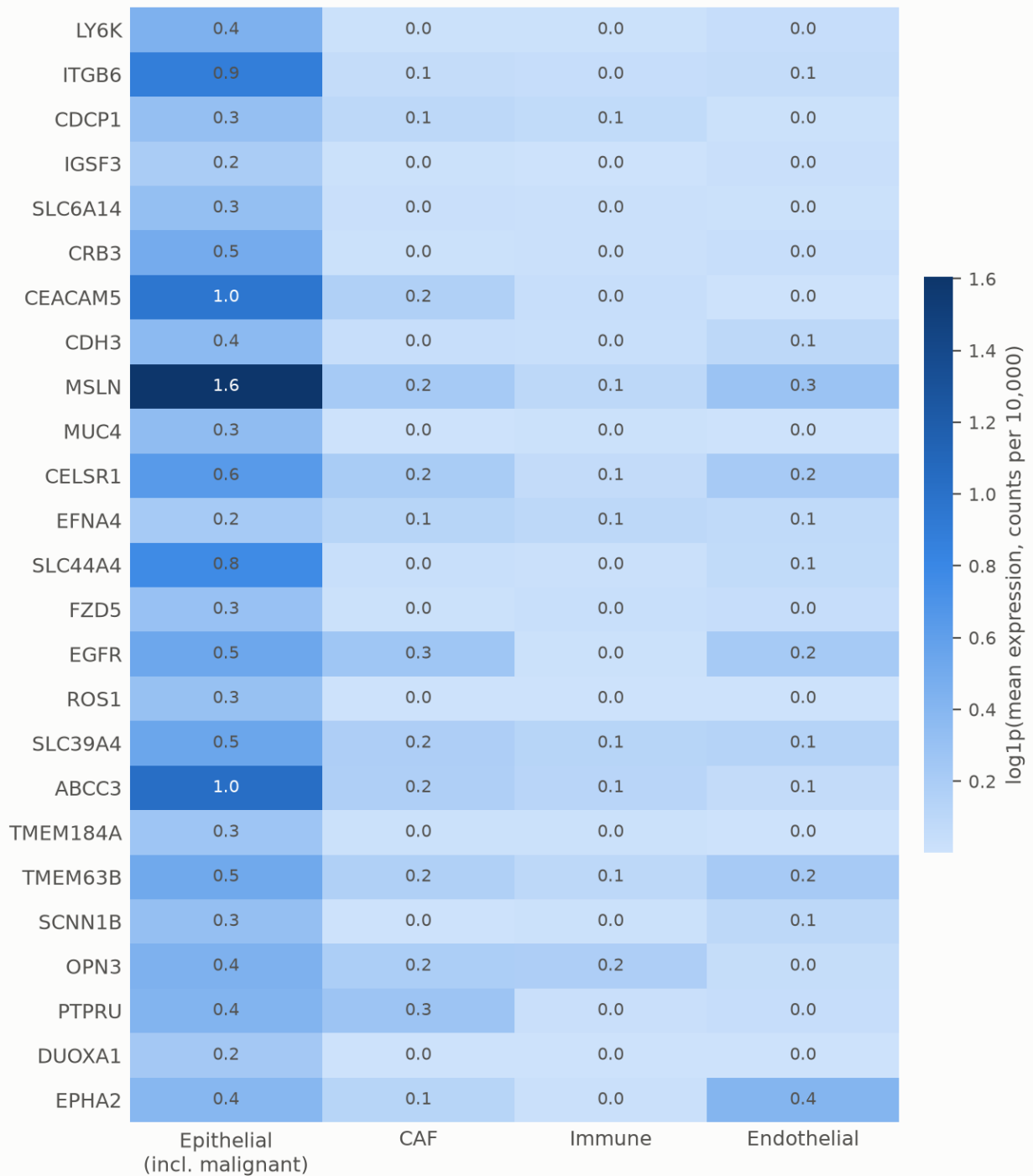
gene	evidence	indication papers	modality papers	drugs in Open Targets
LY6K	10 papers pairing the gene with an antibody/CAR modality	19	10	-
ITGB6	4 drug programmes in Open Targets (max stage n/a), 21 surface-modality	30	21	ABITUZUMAB; STX-100; SIGVOTATUG VEDOTIN ...
CDCP1	70 papers pairing the gene with an antibody/CAR modality	67	70	-
IGSF3	5 indication papers, 2 modality papers	5	2	-
SLC6A14	12 lung-cancer papers but no established surface-targeting programme	12	1	-
CRB3	11 papers pairing the gene with an antibody/CAR modality	5	11	-
CEACAM5	5 drug programmes in Open Targets (max stage n/a), 147 surface-modality	111	147	ARCITUMOMAB; LABETUZUMAB ...
CDH3	1 drug programmes in Open Targets (max stage n/a), 22 surface-modality	42	22	PF-03732010
MSLN	5 drug programmes in Open Targets (max stage n/a), 295 surface-modality	100	295	SS1(DSFV)-PE38; ANETUMAB RAVTANSINE ...
MUC4	157 papers pairing the gene with an antibody/CAR modality	102	157	-
CELSR1	5 papers pairing the gene with an antibody/CAR modality	5	5	-
EFNA4	9 papers pairing the gene with an antibody/CAR modality	12	9	-

Figures

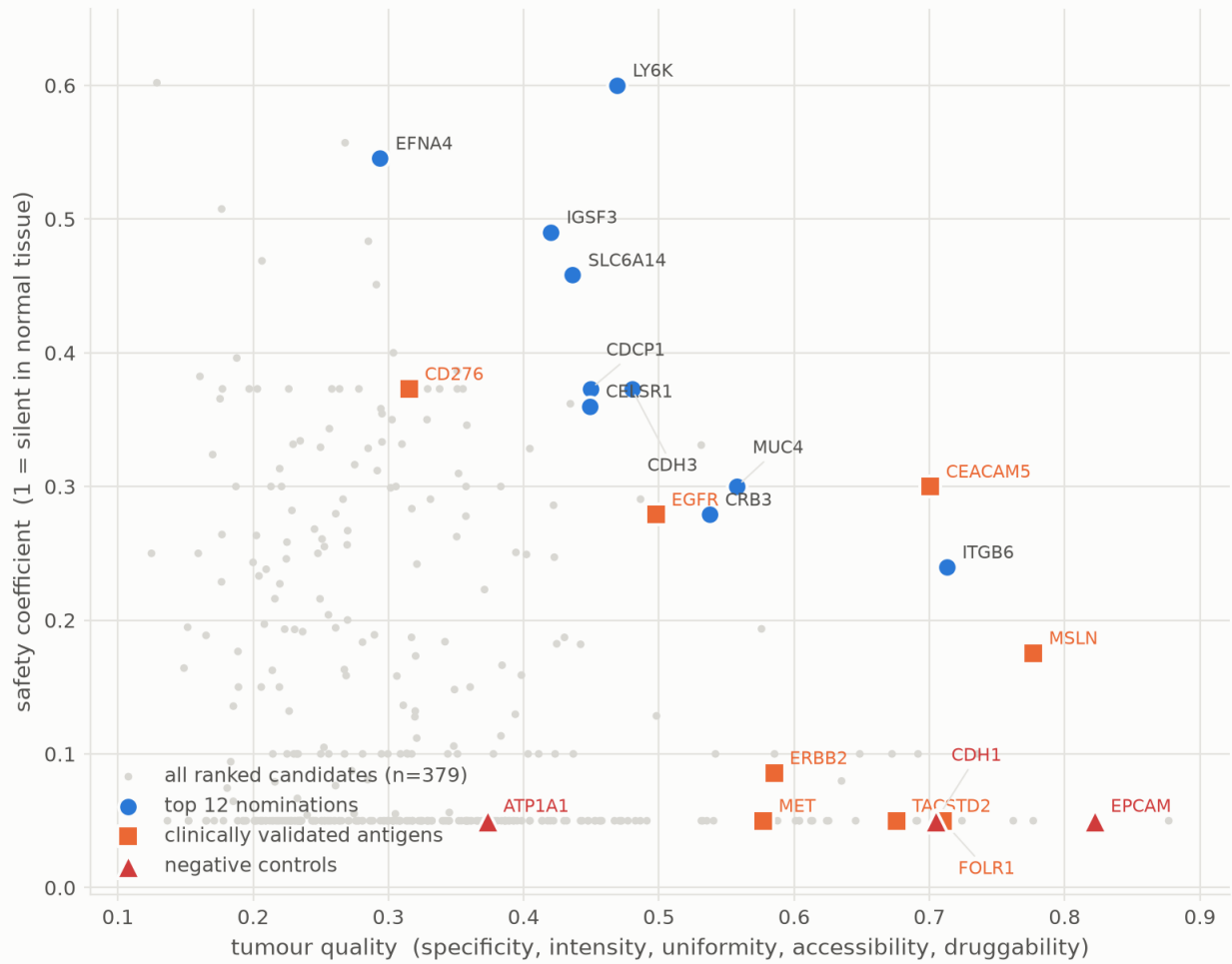
Four figures were written as PNG and SVG. The export bundle passed 8/8 consistency gates, which cross-check the cohort cell accounting against the expression matrix, the reported recall string against the per-gene ranks, the negative-control verdicts against the rule, and the literature table against the ranking.

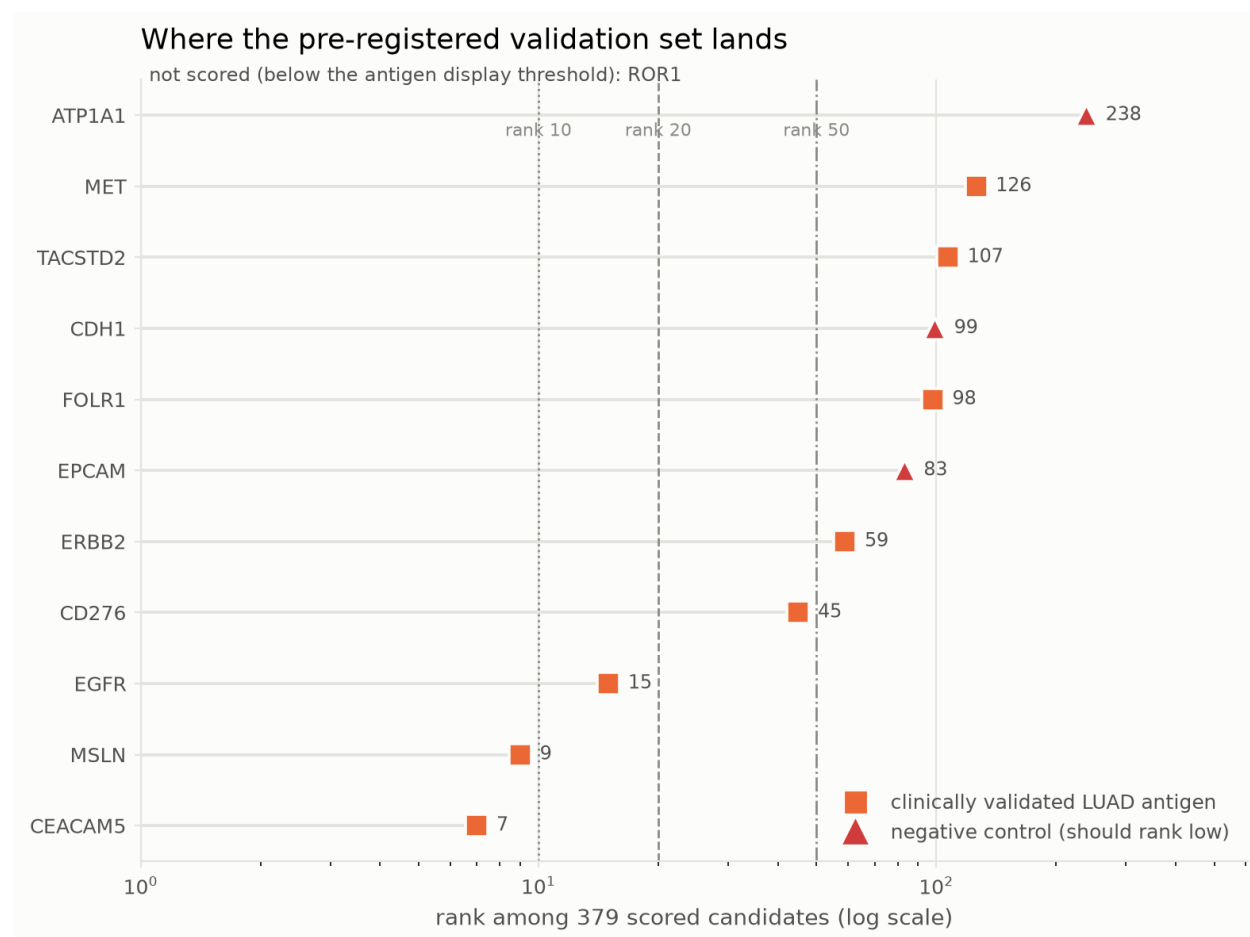


Compartment expression of the top candidates
cross-atlas consensus, counts per 10,000 per cell



Therapeutic index: tumour profile against normal-tissue risk





Export gates

consistency gate	result	detail
cohort cell accounting	PASS	75,095 cells summed over compartments = 75,095 in the cohort table = 75,095 in provenance
sampling ceiling	PASS	max sampled per atlas 20,000 <= ceiling 20,000
expression matrix coverage	PASS	2,674/2,674 genes carry statistics from all 5 atlases
ranking integrity	PASS	379 unique genes, scores monotone in [0,1], range 0.151-0.633
recall report	PASS	@10: reported 2/9, recomputed 2/9; @20: reported 3/9, recomputed 3/9
negative-control verdicts	PASS	EPCAM=PASS(rank 83); CDH1=PASS(rank 99); ATP1A1=PASS(rank 238); CGN=PASS(rank -)
literature coverage	PASS	12 reviewed candidates match ranks 1-12
figures present	PASS	8 files (4 figures x PNG+SVG)

Limitations

- Transcript abundance is a proxy for surface protein density; an ADC needs protein copies per cell, which RNA can over- or under-state.
- Compartment assignment relies on the Cell Ontology labels supplied by each atlas; malignant cells are not re-called from copy-number, so the epithelial compartment mixes malignant and normal epithelium.
- HPA immunohistochemistry is antibody-dependent and semi-quantitative; a missing or weak antibody looks like safety.
- The safety model weights organs by how badly antigen-positive killing is tolerated. Those weights are a stated judgement, not a measurement.
- Novelty in the literature triage means no indexed surface-targeting paper, which is not the same as no prior art.

Data sources

SURFY surfaceome: https://wollscheidlab.org/SURFY/table_S3_surfaceome.xlsx

CZ CELLxGENE Census 2025-11-08

Human Protein Atlas consensus RNA: https://www.proteinatlas.org/download/tsv/rna_tissue_consensus.tsv.zip

Human Protein Atlas normal-tissue IHC: https://www.proteinatlas.org/download/tsv/normal_ihc_data.tsv.zip

Open Targets Platform GraphQL: <https://api.platform.opentargets.org/api/v4/graphql>

DepMap CRISPR gene effect: <https://ndownloader.figshare.com/files/43346616>

Europe PMC: <https://www.ebi.ac.uk/europepmc/webservices/rest/search>